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ORIGINAL PAPER

Coworking spaces for remote workers: an inclusive solution? Advantages and challenges from affiliated workers' perspectives

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Abstract
Some companies are considering offering their affiliated workers external coworking spaces as an option to achieve both flexibility and inclusion, giving remote workers the chance to maintain a sense of community and have access to on-site support. However, the motivations, advantages, and problems associated with using coworking spaces for affiliated workers remain underanalysed. This study adopts a qualitative approach by conducting semistructured interviews with Italian affiliated workers who are coworking space users. Our analysis explores the benefits of using coworking spaces and the reasons that motivate affiliated workers to do so, as well as how relationships, social support, and collaboration with coworkers and colleagues are perceived. The results show that the perceived main advantages concern (a) flexibility and work efficiency and (b) relational opportunities, at both the personal and professional levels. The findings clarify the conditions under which workers sharing coworking spaces develop satisfying social relationships and collaboration, shedding light on the group dynamics of a mix of affiliated and unaffiliated workers across coworking spaces from the affiliated workers' perspective. The study also discusses the role that affiliated workers' organizations can play in addressing the challenges that coworking spaces pose for workers' well-being and organizations.

Keywords Coworking spaces · Remote working · Qualitative research · Workplace relationships · Affiliated workers

JEL codes O15 · J24 · J28

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1 Introduction

In organizations, transformations towards postbureaucratic models (Waters-Lynch and Duff 2021; Kraus et al. 2022, 2023) are influencing the configuration and uses of workspaces and changing working practices and organizational and personal relationships (Halford 2005).

In addition, the COVID-19 pandemic has accelerated the process of digitalization and the exploration of new work arrangements, while the numbers of teleworkers and workers from home have rapidly increased (Ker et al. 2021; Miglioretti et al. 2023). All countries for which comparable observations are available experienced an increased rate of teleworking during the COVID-19 pandemic. In Italy, teleworking rates in the second quarter of 2020 were more than 4 times greater than those before the pandemic, increasing by 15% points on an annual basis (Criscuolo et al. 2022). During the same period, the number of coworking spaces also increased around the world despite the pandemic, although many had to temporarily close their doors as various governments issued "stay-at-home" orders (Deskmag 2022). Moreover, there are some predictions that coworking will be even more necessary and prevalent in the postpandemic world, as more companies are shifting to remote work, and thus, more individuals will need alternative workspace options (Howell 2022). Coworking spaces emerged as a new and promising phenomenon in the early 2000s as a new category of flexible workspace for entrepreneurs and freelancers and have been defined by some researchers as a type of *third place* (Oldenburg and Brissett 1982) due to characteristics that situate them between private homes and corporate offices (Brinks 2022).

Previous literature regarding coworking spaces has focused predominantly on the motivations, advantages, and problems for *unaffiliated* workers, such as entrepreneurs and freelancers, in using such spaces, while the use of these spaces by *affiliated* workers, such as remote workers, has been underanalysed. Fewer investigations have considered the use of coworking spaces by affiliated workers (Josef 2017; Walden 2019; Jeske and Ruwe 2019) or teleworkers (Lescarret et al. 2022; Rodighiero 2023). This gap in the literature appears significant to address, especially in light of the emerging trends in the use of coworking space. According to the Global Coworking Survey 2019, one-third of coworking users are affiliated workers who telework, and one of four users has his or her membership fee paid by an employer or client (Foertsch 2019).

Following the COVID-19 pandemic, both large and small companies have had to reevaluate their workspace plans. Many have recently announced plans to become "all-remote" organizations by having no central office or headquarters and allowing their affiliated workers to work wherever they want. Other companies plan to reduce their number of offices and favour working from home or other places. As part of this plan, some companies are considering offering their affiliated workers a coworking space or a certain amount of money each month to purchase a coworking space in which affiliated workers who do not want to work from home are able to work (CNBC 2021). Spreitzer et al. (2015) indicated tapping into new ideas, flexibility, and autonomy as the major reasons for companies to favour the use of coworking spaces among their affiliated workers. In addition, Jeske and Ruwe (2019) proposed cowork-

ing as an option to achieve both flexibility and inclusion, giving remote workers the chance to maintain a sense of community and access to on-site support. In fact, some research has shown that relational dynamics that are established in coworking spaces can positively influence collaboration, productivity and well-being (Bueno et al. 2018). Nevertheless, as Appel-Meulenbroek et al. 2021 noted, users' perception of the benefits of coworking, even if the decision to use coworking spaces is based on company policies, strongly depends on the users' expectations and preferences. To understand whether and how coworking can be advantageous for affiliated workers, it is crucial to investigate the experiences of affiliated workers who use coworking spaces.

In this work, we are interested in analysing the perspectives of affiliated workers on using coworking spaces. We explore the benefits and reasons that motivate workers to use coworking spaces and how relationships, social support and collaboration are perceived with respect to both coworkers who share the space and colleagues who work elsewhere.

We believe that having an affiliation with the company, according to organizational identification theory and the professional identity perspective (Ashforth and Mael 1989), can impact how individuals experience coworking in terms of both work practices and social relationships.

We are interested in highlighting the perceived advantages of coworking spaces for affiliated workers and how their organizations can influence their work experience in coworking. We also aim to clarify the conditions under which these workers develop social relationships and collaboration and the differences between social relationships with other coworking space users and colleagues in their organizations who do not use the same spaces. Thus, our study contributes to the literature on coworking spaces by shedding light on the group dynamics of a mix of affiliated and unaffiliated workers across coworking spaces from the perspective of affiliated workers.

1.1 Current trends in coworking spaces: characteristics and evolution

Coworking spaces can be defined as subscription-based workspaces in which individuals and teams from different companies work in a shared, communal space (Howell 2022). They encompass various forms of contemporary open workspaces that provide shared office facilities and infrastructure (Wi-Fi, electricity, printing services) to people from highly diverse professional backgrounds, including freelancers, entrepreneurs, startups, microenterprises, and, increasingly, workers affiliated with companies and corporate teams (Waters-Lynch et al. 2016; Bouncken and Reuschl 2018; Mitev et al. 2019; Schäfer et al. 2023). In fact, coworking spaces were originally developed as open offices targeting independent professionals who were attracted by the idea of participating in a collaborative work environment. In recent years, companies have not only invested in the design of mostly internal coworking spaces but also rented desks from independent coworking spaces for their affiliated workers and teleworkers (Spreitzer et al. 2015; Bouncken et al. 2021). Josef (2020, pp. 66), including the perspective of companies that use coworking space not as a primary but as an additional work arrangement, proposed the following definition: "*Coworking spaces are neutral places, owned and operated by a private or legal entity, where*

affiliated and unaffiliated professionals work side by side or in collaboration. The spaces are used by individuals, teams or other cross-organisational groups during a specific project phase or for an unlimited period, in addition to other work scenarios or exclusively". This phenomenon is most frequent in large cities in Asia and North America, while freelancers continue to be the main users in Europe and in small and medium-sized cities in general (Foertsch 2020). In the Italian context, the distribution of coworking spaces varies widely across the country. The few data available seem to confirm this trend: Italian coworking spaces seem to be more populated by freelancers, followed by companies and startups, private sector-affiliated workers, and a smaller number of public sector employees who work remotely (Italian Coworking 2022). Coworking spaces are becoming increasingly widespread and constantly evolving. They generally promote themselves as "membership communities" with sociality and the potential for meaningful relationships between coworkers as a key value (Waters-Lynch et al. 2016). This is pursued through the physical environment; such spaces usually have different workstations, such as flexible desks in open-plan offices (on a "first-come, first-served" basis), fixed desks and private offices that offer space for small teams and, in some cases, meeting rooms. In addition, they offer areas for drinking coffee, eating and relaxing. Nevertheless, they vary significantly regarding equipment, philosophy, design, size, clients, services, ambient conditions and rules. Kojo and Nenonen (2016) differentiated coworking space based on the business model (for-profit/nonprofit) and the level of access for users (public/semipublic/private). Spinuzzi (2012) described different types of coworking spaces based on the level of collaboration between users. Orel and Bennis (2021), in their recent classification efforts, distinguished between the traditional type of coworking space known as "*individual-purposed*" spaces, where freelancers, remote workers, and other location-independent professionals gather in an open-plan office environment, and the newer concept of *"group-purposed"* spaces. The latter focus on teams rather than individuals, aiming to connect both individuals and teams across different institutions. However, although some companies have been attracted by the idea of fostering collaboration and cocreation with external stakeholders, in most cases, the primary focus is flexible office space, and community facilitation and interinstitutional interactions are often minimal. Informal networking mechanisms are generally minimized and replaced by more formal interactions; thus, to maintain the tenant company's preexisting company culture, they have lavish facilities compared to traditional coworking spaces (Bréchignac et al. 2016). The literature therefore highlights that, depending on their objectives and philosophies, coworking spaces support social interactions in different ways. Environments vary in terms of how actively they do so, the depth of the relationships and the extent of the collaboration, i.e., from mere watercooler talk to active knowledge and expertise sharing and business partnerships (Orel and Bennis 2021).

1.2 Reasons for using coworking spaces: the point of view of companies and workers

Whether and how coworking spaces can benefit companies at the organizational level have not been extensively studied, although researchers have discussed how their use

can change work and company management practices (Leclercq-Vandelannoitte and Isaac 2016). Only recently have some scholars begun to explore the advantages of offering remote workers or teleworkers a more supportive environment than home offices for productivity and well-being based on previous research on self-employed and freelance workers' motivations for using coworking spaces (Josef 2020; Berbegal-Mirabent 2021; Manzini Ceinar and Mariotti 2022). Similarly, companies are increasingly using coworking spaces to facilitate internal interactions, cross-team collaboration, and innovation, boost productivity, and expose their affiliated workers to new talent and expertise beyond their boundaries (Sargent et al. 2018; Bouncken et al. 2021). Additionally, this approach is perceived to benefit remote workers by exploiting the social characteristics of coworking space to reduce the risk of social and professional isolation (Bartel et al. 2012), facilitating relationships between coworkers or colleagues. In addition, the mix of structure and the flexibility of coworking (Josef 2020), together with the availability of adequate technology and work equipment, could favour workers' work–life balance. Coworking can provide spatial flexibility, which helps workers achieve mobility and structure in their work routines. Especially for workers who have segmentation preferences (Kreiner 2006) in managing the boundary between work and life, coworking is an appealing option, offering flexibility and a manageable commute while reducing the risk of feeling "always connected", which is a common experience when working from home (Byron 2005). For companies, coworking spaces can also represent an opportunity to save on costs related to office rent; thus, this approach can promote well-being and productivity for both companies and workers through flexibility (Manzini Ceinar and Mariotti 2022). However, Leclercq-Vandelannoitte (2021) discussed the contradictions in the use of coworking spaces by companies, which can adopt these spaces for teleworkers in an effort to "rematerialize" the organization and exercise control over affiliated workers (by, for example, monitoring employees during work hours).

The literature exploring the perspectives of affiliated and remote workers seems to be limited, but recent studies have indicated that the motives and benefits of affiliated workers may somewhat differ from those of self-employed or other categories of workers in coworking spaces and may be influenced by organizational factors of their employer, as well as by countries (Appel-Meulenbroek et al. 2021).

Previous research has shown that freelancers use coworking not only for economic reasons (saving on office rent costs) but also for reducing isolation, finding a sense of community and creating a professional network or occasions for knowledge exchange for personal and professional advantage (Spinuzzi 2012; Gandini 2015; Rådman et al. 2023). In addition, coworking users often cite the benefits of aspects related to coworking space efficiency (e.g., increased creativity, productivity, motivation, and structure) and flexibility in terms of time and infrastructure (Servaty et al. 2018; Appel-Meulenbroek et al. 2021). Coworking spaces are also perceived as having a positive effect on well-being compared to home offices in terms of concentration, productivity, self-organization, social interaction, and separation of private and work life, as well as on overall satisfaction (Robelski et al. 2019; Servaty et al. 2018). Reuschke et al. (2021) emphasize that coworking empowers freelancers and entrepreneurs to establish routines, enhancing both productivity and a sense of accomplishment. This in turn aids in shaping a positive professional identity.

Bouncken et al. (2020) explored novel avenues in the literature, employing an institutional theory perspective (DiMaggio & Powell, 1983) to investigate institutional patterns within coworking spaces and elucidate their correlations with work satisfaction. Institutional patterns arise from both the formal and informal organization of the space and the conduct of its users. Over time, these behaviours are collectively adopted and institutionalized, influencing conduct in the space. The study revealed that institutional patterns in coworking are linked to social capital, participation, autonomy, linkage multiplicity, and collaborative knowledge creation, with individual work satisfaction as the resultant variable. Unfortunately, Bouncken et al. (2020) did not distinguish between affiliated and unaffiliated workers, and how affiliated workers do or do not participate in the definition of institutional patterns, or in the dynamics of these patterns of their organizations, must be explored.

For example, a recent study exclusively on teleworkers revealed that the intention to use coworking space for telework is also strongly influenced by the cost of membership and concerns about increased commuting time (Lescarret et al. 2022). It remains to be clarified whether these results are valid for those who have experience with coworking, given that the participants in that study were not current coworking space users. Furthermore, Leclercq-Vandelannoitte (2021) found contrasting effects on workers' experience in response to an organizational policy of maintaining coworking spaces for teleworkers; for some workers, the spaces recreated opportunities to signal their presence and engagement, but for others, they introduced new constraints of availability, exposure, and visibility, highlighting that the subjective experience in coworking should be understood through organizational practices and psychological bonds with affiliated workers' companies and, consequently, with colleagues.

Overall, the current literature reveals not only the perceived advantages of coworking spaces but also the differences between affiliated workers with flexible work arrangements and traditional coworking users, highlighting the importance of research illuminating their specific working conditions.

We thus propose the following research questions in this study: what are the reasons for using coworking spaces, and what are the perceived benefits of coworking spaces for affiliated workers? How do companies support or undermine their affiliated workers' experience in coworking?

1.3 Collaborative and supportive relationships in coworking spaces

As noted, the social dimension of coworking spaces attracts companies and workers. Thus, now that such spaces are increasingly populated by affiliated workers, as (Ross and Ressia 2015) argued, it is interesting to investigate how their presence can change the observed social dynamics, creating a mix of affiliated and unaffiliated workers across the coworking spaces.

The literature has shown that coworking spaces traditionally promote the values of collaboration, exchange, and community (Ivaldi et al. 2022). Many intentionally bring people together to develop collaborative relationships, social learning (Bilandzic and Foth 2013) and social support (Bianchi et al. 2018), thus sustaining a sense of community at work (Garrett et al. 2017).

According to previous research, social interactions between coworking space users come in various forms (Gerdenitsch et al. 2016), fulfilling both personal and professional needs. Scientific research has demonstrated that in coworking spaces, the most frequent type of interaction seems to be informal communication, such as conversations over coffee and lunch, which can be more or less intimate, but coworkers can also experience social support (Deskmag 2013; Gerdenitsch et al. 2016). It has been observed that coworking users sustain each other through "instrumental support", such as the "exchange of information, sharing knowledge and advice" or even moral support. Coworking users also collaborate, creating synergies or common benefits between their own businesses, even if rarely on the same project.

Thus, several authors have shown that coworking spaces encourage cooperation among coworking members (Capdevila 2013; Waters-Lynch et al. 2016; Orel and Almeida 2019) and that the perception of social support has professional benefits that are positively related to performance satisfaction (Gerdenitsch et al. 2016).

Since the emergence of coworking spaces, scholars have sought to describe and understand the mechanisms underlying social relations, showing the role of physical spaces (Bouncken et al. 2021), coworking management practices (Parrino 2015), and collective actions that create a sense of community and atmosphere that facilitates supportive relationships (Garrett et al. 2017; Orel and Almeida 2019). It has also been shown that coworking spaces have different degrees of "community potential" (Berdicchia et al. 2023), and if they are not well managed, users can experience social isolation and stress (Bouncken et al. 2018). Using a sociomaterial perspective that reflects the importance of the interconnectedness of social and material elements (e.g., spatial design, physical layouts, colour schemes, shared infrastructure and less visible artefacts such as information systems), common social spaces within coworking spaces, such as cafeterias, lounges and bars, have been found to be key in fostering a collaborative community (Bouncken et al. 2021) and a "positive ambience" (Orel and Almeida 2019).

Others have highlighted that social relations are typically enabled through a variety of "organizational platforms" (Parrino 2015), ranging from internal digital social networking sites and social events to community hosts or managers, that facilitate personal introductions and social networks. It has also been shown that these devices and practices are the basis of identification with both the coworking community and the coworking space itself, defined as the "sense of community", which is considered a core mechanism in the coworking space to achieve a high frequency of collaboration and supportive interactions (Garrett et al. 2017; Waters-Lynch and Potts 2017; Spinuzzi et al. 2019). However, Parrino noted that coworking spaces differ on a continuum with the "presence of an articulated platform of tools and initiatives designed to stimulate interaction and collaboration" and a "total absence of such offers" (p. 5).

Thus, research highlights that processes of organizational identification, formed based on social interactions, play a crucial role in fostering collaboration. Workplace identification indicates the degree to which an individual sees himself or herself as a member of a workplace and incorporates the organization into his or her self-concept (Ashforth and Mael 1989). More specifically, identification based on coworking is proposed to emerge from emotional responses to the coworking space and its members, as individuals cultivate a sense of community (Spreitzer et al. 2020).

Little research exists on how these social dynamics are experienced by affiliated workers using coworking spaces inhabited by affiliated and unaffiliated workers.

For example, the increased presence of remote workers may introduce challenges related to both territorial aspects, such as differences in how remote workers and freelancers or entrepreneurs utilize artefacts and workspaces, and atmospheric factors, such as tensions arising from varying preferences for productive versus socializing environments within collaborative workspaces (Rodighiero 2023).

There are conflicting results regarding the social relationships among affiliated workers in coworking spaces. It has been suggested that positive informal relationships and collaboration may occur but, at the same time, can be hindered by psychological factors or coworking characteristics.

On the one hand, remote workers, particularly high-intensity teleworkers, could be motivated to build relationships in coworking spaces since, similarly to freelancers, they may experience professional isolation (Golden et al. 2008). In fact, it has been demonstrated that teleworkers expand their professional networks and receive and ask for support from other coworking-space members (Walden 2019). On the other hand, the same research based on the identity theory perspective also showed that teleworkers may experience role conflict when they receive requests for help from members who do not work for the same company, perceiving them as disturbing their work. In Josef's research (2020), although respondents recognized contact with other workers as beneficial, they did not have close relationships with other workers in the coworking space. In this case, the affiliated workers had only been using the coworking space for a few months and then only rarely. In addition, according to established evidence on identity conflicts at work (Horton et al. 2014), it can be assumed that a sense of belonging to one's own organization can hinder identification with the coworking community and therefore establishing collaborative relationships. Dual identification with both the company and the coworking space can make it more difficult for remote workers to intertwine their own professional identity with those of other coworking-space members and participate in constructing a collective coworking identity. Spreitzer and colleagues (2020) also suggest that coworking-based identification is lower than company-based identification for some remote workers.

According to a recent study, many remote workers do not demonstrate a strong sense of identification with the coworking space they use. Nevertheless, the study underscored the significance of social support as a counterbalance to this lack of identification, contributing to the well-being of remote workers (Rodighiero 2023). Depending on the type of work, affiliated workers may look less for external professional networks, or they may rely on some sort of support from their colleagues (Collins et al. 2016). Furthermore, the collaboration that occurs when there is a perception of reciprocity in the exchange may be prevented if the necessary expertise is lacking among coworking-space members. It has also been noted that some coworking communities may not provide the amount of social support expected when their members change frequently (Jeske and Ruwe 2019).

However, the influence of these aspects on building connections with coworking members and on the experience of collaborative behaviours in coworking spaces has not been extensively investigated.

Thus, considering the above gaps in the literature, we also aim to answer the following research questions regarding how affiliated workers use coworking spaces: how are social relationships perceived between coworking users? Are social support and collaboration experienced? Does being affiliated with and having colleagues inside or outside the coworking space affect coworking relationships?

2 Methodology

This study adopted a qualitative approach by conducting semistructured interviews with coworking-space users to access the participants' interpretations, perceptions, and experiences to gain an understanding of their world, in line with previous research in the field (Orel and Mayerhoffer 2021).

We conducted a national study on coworking spaces in Italy. Statistics show a growing use of these spaces and the adoption of teleworking after the COVID-19 pandemic.

The participants were identified through a purposive sampling strategy involving people who were using a coworking space and who were affiliated workers of a company (we interviewed volunteers who identified as affiliated workers with an employment contract with an organization). In addition, community managers in charge of the organizational and social aspects of coworking spaces were included to provide a complementary perspective. To recruit participants, the research was disseminated to the community managers of various coworking spaces by an email, initially sent to personal contacts of the researchers, that explained the study's goals in detail. Community managers interested in the research forwarded the participation proposal to affiliated workers who used their coworking space. The total sample was composed of 17 participants; a total of 14 coworkers and 3 community managers participated in the study. Table 1 reports the participants' main sociodemographic characteristics. Regarding the job characteristics of the affiliated workers working in coworking spaces, 7 were fully remote workers with all their colleagues working outside of their coworking space, 2 of whom were affiliated workers of a company based abroad. Six participants worked in the coworking space with either some colleagues ($n=3$) or their entire company, which was in most cases a startup ($n=3$). Overall, the participants came from 9 coworking spaces located in different cities in Italy. Five of the participants worked in the information technology or consulting sector, and the other common sectors were graphic or web design, media or journalism, and marketing. Finally, 9 participants stated that they worked for a small company, 3 worked for a medium-sized company, and 3 worked for a large company. Most of the coworking users were male ($n=11$) and aged 22 to 58 years; the majority lived with at least 1 other person, and 7 had children. Most used a coworking space close to their home (less than 25 minutes' distance), except for 3 participants, 2 of whose companies were located in the coworking space. The participants were assigned pseudonyms, and the names of the coworking spaces were changed for confidentiality. These two pieces of information are referenced in the quotes.

The interviews were conducted between June and September 2021 by videoconference (Skype9); in most cases, the interviewees were in their coworking space,

Table 1 Participants

Name	Gender	Age	Cohabitants	Job role	Company size	Month in CW	Days in CW/week	Distance from home (minutes)	CW	Colleagues inside or outside the CW
Andrew	M	40	3	Employee of information company	Small	9	5	15	AA	Inside All the company members
Camilla	W	29	1	Employee of publishing company	Large	24	2	20	BB	Inside Some of the company members
Daniel	M	27	3	Technical assistance manager	Medium	12	6	20	CC	Outside and inside
Frances	W	36	/	E-mail marketing consultant	Small	60	5	10	DD	Outside
Giulie	W	58	/	Sale director	Medium	24	5	10	EE	Outside and inside
Max	M	40	2	UI/UX designer/Front-End Developer	Small	48	3	20	EE	Outside
Luke	M	38	1	Office worker	Large	10	2	16	FF	Outside
Monica	W	35	3	Press office	Small	3	4	15	GG	Outside
Orlando	M	33	1	Social media manager	Small	3	5	30	EE	Inside All the company members
Paul	M	22	4	Customer service officer	Small	2	5	45	HH	Inside All the company members
Peter	M	37	2	Engineer	Large	24	2	15	EE	Outside
Ricky	M	30	/	Product and interior designer	Large	2	3	5	EE	Outside
Stephan	M	36	1	Digital product designer	Small	12	4	5	II	Inside All the company members
Teo	M	37	3	Computer consultant	Medium	7	4	30	EE	Outside
Ellis	F	35	1	Community manager	Small	36	5	15		
Simona	F	25	/	Community manager	Small	7	5	15		
Jane	F	33	1	Community manager	Small	62	5	12		

allowing some to show part of the space via their webcams, while in others, they were at home or in other locations (e.g., public gardens). Each interview lasted 60 min on average. An interview schedule was used, but the process was flexible and allowed the participants to introduce issues they felt important. The interview guide for coworkers was divided into two parts according to the research questions. The first part addressed the motives and benefits of working in a coworking space. Furthermore, we asked for a description of the coworking space and the participants' daily work time organization/routine. The second part included questions about perceived social relationships in the coworking space and with office-based colleagues. In conclusion, participants were asked about their general satisfaction with their work experience in the coworking space. Community managers were asked about the reasons for and benefits of affiliated workers using coworking spaces and about social relationships in those spaces.

2.1 Data analysis

The data were analysed using thematic analysis (Braun and Clarke 2006) and managed in NVivo version 11. The relevant material was selected and subsequently encoded according to themes emerging from the dataset. The identified themes captured important aspects of the data in relation to the research questions. The thematic analysis was conducted through dialectical interaction between deductive and inductive reasoning. During this process, attention was given to new emerging aspects in the texts; therefore, new categories could be constructed until a final configuration of the themes was reached. The data were categorized into key themes and then further refined into subthemes. Systematic data interpretation was guided by the principles of the constant comparative method (Corbin and Strauss 2008), which involves continuously comparing interviews and categories, i.e., comparing interviews with each other, comparing categories with each other, and comparing all the interviews and categories with theoretical references. The data were thus organized in central thematic areas that were complementary and transversal to the analytic categories. The narrative writing of the central thematic areas was a reflexive process that implied redefinition of the final configuration of the findings aimed at producing a coherent "story". Thus, rival configurations of themes were ultimately modified (Patton 2002), and quotes were selected that best illustrated themes from the participants' perspective. The credibility of the analysis, as a criterion for qualitative research, was assessed through supervised sessions to check the coding strategies and review the interpretation of the data by discussing any reason for variation (Barbour 2001). All the participants provided written informed consent and approved the use of anonymous data for publication. Ethical approval was obtained from the Ethics Commission of the Psychology Department of the University of Milano-Bicocca (RM-2021-433). The study is in accordance with the Helsinki Declaration of 1964.

3 Results

To answer the research questions, the results were organized into two main themes. The first *theme*, *coworking as a supportive work environment*, describes the reasons for and benefits of coworking and companies' support for it, while the second *theme*, *building relationships*, reports the nature of the relationships with colleagues who work outside the coworking space and with coworking users and colleagues in the space. Figure 1 represents the themes identified through the thematic analysis, which will be described in the next sections.

3.1 Coworking spaces as a supportive work environment

The respondents were generally very satisfied with their work experience in the coworking spaces; in several cases, they did not list any disadvantages, were not willing to change work situations, and expressed dissatisfaction with fully remote work. They valued the physical and social aspects of coworking spaces and the effects of their use on well-being, productivity, and flexibility.

Additionally, the role of expectations that some individuals demonstrated towards the organization emerged. However, excluding the cases in which a whole company was present in the coworking space, there were only three cases in which the use of the space was promoted by a company that paid the rent; for the other workers, coworking was an individual alternative to working at home.

The following sections describe the reasons for and benefits of using coworking, which fall into two main categories: *coworking as a flexible and efficient solution* and *coworking as a relationship catalyst*.

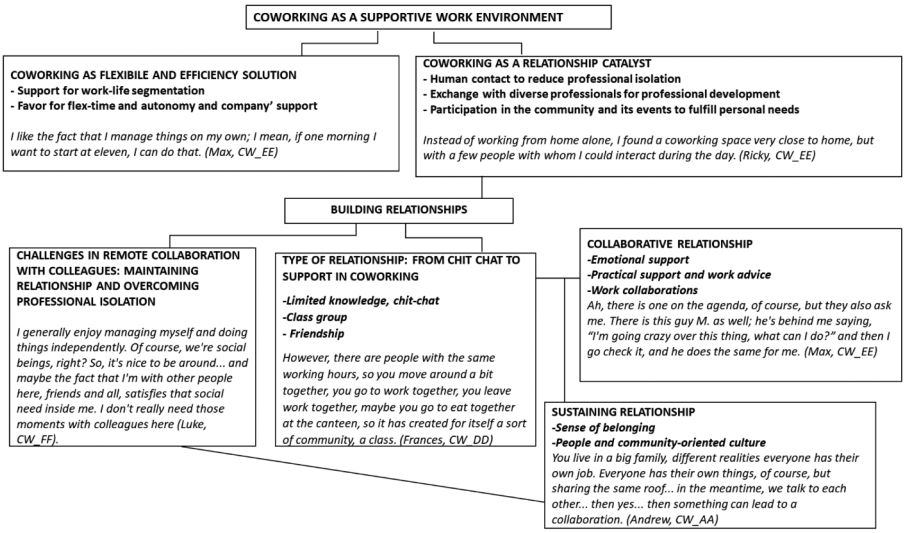

Fig. 1 Themes and subcategories: using coworking spaces from the affiliated workers' perspective

3.1.1 Coworking as a flexible and efficient solution

The coworking space was described by more than half of the affiliated workers as an environment chosen or appreciated for its advantages in favour of work–life balance, work well-being and job performance. They frequently expressed feeling more focused, motivated, and satisfied with their daily job activities than when they worked from home. This section explores this theme and its subcategories.

Support for work–life segmentation. The use of coworking space was positively described as supporting segmentation for work–life balance, which was considered important for work well-being, as opposed to the working from home experienced by many during the pandemic or before their use of coworking space. The positive aspects related to having a dedicated place to work other than the home or closer to home than employers' offices for remote workers. In addition, the physical and infrastructural aspects of the space were recognized as positive. Carrying out work activities outside the home allows the demarcation between the two spheres and confines work to an external place "*where you can concentrate and give your best*" (Daniel, CW_CC). This approach helps to reduce the overlap between domestic and work activities or interruptions felt to hinder concentration when space is shared with family members. For example, an interviewee described "*destructive multitasking*" when working from home as follows:

> I do not hide the fact that while I work from home, I run the dishwasher, the washing machine. I take ten minutes and wash the floor… it is damaging… it is damaging. In fact, I hate working from home; I always want to go to the office… (Camilla, CW_BB).

The coworking space could benefit concentration and perceived productivity, also favouring the segmentation of time in contrast to constant connectivity and "*giving office rhythms*", as Max explained:

> On the other hand, a coworking space is cool because it almost gives an office rhythm, which for me is a big advantage in the sense that I have my work brackets in which I have to concentrate absolutely; therefore, it psychologically makes me cut out many distractions. (Max, CW_EE)

The use of a coworking space allows the performance of rituals that are lost when working from home and that facilitates the assumption of one's work identity, such as changing clothes, applying makeup or commuting. These rituals were seen as vitalizing ways to change roles and prepare for the work day. This was the case for Monica, who, when the interview occurred, could not go to her coworking space or her corporate office, which was located in another Italian region, due to COVID-19 emergency restrictions:

> The thing I miss the most is getting ready and leaving home and going to work […] And I miss the fact of putting on makeup, of beautifying myself a little

> in the morning. Yes, I must say, I miss the detachment from mom to working mom. (Monica, CW_GG)

Another positive element of coworking space was that is often better equipped than workers' homes, with pleasant spaces and additional services, such as canteens, break areas or meeting rooms, that make it a "*a nice workplace to work in*" (Andrew, CW_AA).

> There are also some great services because regardless, you have to eat, you get free water and you can get packages delivered there as there is a reception centre… it is all so convenient. (Paul, CW_HH)

The only mentioned downside was derived from coworkers in an open-space work area. Even if the space was appreciated for helping increase socialization, the interviewees reported a risk of noise or interruptions and lack of privacy as critical issues. However, some participants did not seem to perceive these as major problems; instead, they adopted "isolation" strategies such as wearing headphones, changing locations or, on some occasions, working from home or conference rooms.

Favour for flex-time and autonomy and company support. A few interviewees mentioned the perception of flexibility and autonomy in working hours and workplace location, in response to personal and professional needs, as a positive aspect of coworking spaces. This appeared to be influenced partially by company policies regarding time flexibility and partially by coworking routines. Generally, the interviewees reported having a high degree of time flexibility but tended to maintain traditional office hours to accommodate communication needs with colleagues or customers. They varied only slightly in regard to their times of entry and exit. Others valued flexibility in the choice to work from a coworking space or from home, and the habit of working from home on Fridays appeared widespread. On the one hand, flexibility is supported by the coworking environment being perceived as freer and less controlling than traditional offices.

> To me the cool thing is that here, it's like we are colleagues, but actually everyone does a different job, so you can also do half a day and leave; it is not as if someone will say, "Ah but why did you leave half a day earlier?" Anyway, I like the fact that I manage things on my own; I mean, if one morning I want to start at eleven, I can do that. (Max, CW_EE)

On the other hand, flexibility is possible thanks to company policies that allowed it and that did not control schedules and workplaces. A participant described the culture of his company, which was located in the coworking space, in this way:

> It truly has free "smart working" aspects. That is, one can easily stay at home, just as you please. It has flexible entry times; that is, you are supposed to start at 9, but you can get there at 10 whenever you want… super easy. (Andrew, CW_AA)

However, remote workers reported having different levels of support from their companies when using the coworking space. In two cases, companies promoted the use of the space by paying the rent for their employees' desks; in another case, there was only formal recognition of such use. Many others stated that their companies did not know that they were working in a coworking space. Thus, failure by employers to pay for and high costs of some coworking spaces emerged as obstacles to the use of these spaces in the participants' view, affecting their job satisfaction. For example, Monica stopped using her coworking space when her company stopped renting her desk. Another interviewee criticized his company's policies for remote workers, which did not support coworking spaces. He perceived support for such spaces as a measure promoting workers' well-being:

> As I told you before, I have been commuting for many years, and I think that if my company took over an office like this, it would make our lives much easier […] The last thing I will tell you, for example, during the pandemic, many companies decided to give office chairs to their affiliated workers … now my company is giving us credit for equipping our home in this way. In my opinion, they could have paid for desks in a coworking space alternative. (Daniel, CW_CC)

3.1.2 Coworking as a relationship catalyst

All the interviewees described coworking as offering relational advantages. Three dimensions were more frequently mentioned: (1) *human contact to reduce professional isolation*, (2) *exchange with diverse professionals for professional development* and (3) *participation in the community and its events to fulfil personal needs* as providing benefits at both the personal and work levels. Social advantage was given as the main motivation for using such a space, while for some, such use determined their choice of coworking space.

Human contact to reduce professional isolation. In contrast to working from home, for many participants, the coworking space satisfied their social needs and made their work day more pleasant, which had a direct effect on their well-being and motivation. Their choice to use such a space was motivated by the need to enhance social contact and reduce social isolation, which was exacerbated in some cases by the lack of informal contact with remote colleagues. An example was given by Ricky:

> Instead of working from home alone, I found a coworking space very close to home, but with a few people with whom I could interact during the day. (Ricky, CW_EE)

Having someone just to chat and share breaks with was described as a positive characteristic of the coworking environment that makes workdays seem shorter and more enjoyable than those experienced at home.

> So, like, imagine you go grab lunch with a coworker or share a laugh, just chatting with someone. It's way nicer, you know? Makes the day fly by, 'cause otherwise, being stuck at home, the day just drags on. (Daniel, CW_CC)

This advantage motivated the choice to go to a coworking space for those who had flexibility in choosing days to work from home but appeared more pronounced for those who had experienced fully remote work.

Exchange with diverse professionals for professional development. For many interviewees, professional diversity was the main positive characteristic of coworking. The participants reported appreciating the opportunity to meet people from different work environments and professional backgrounds. However, one participant perceived the coworking space to be more suitable for startup teams. For some, using the coworking space was seen as an occasion for individual professional growth, which was rare in a traditional office. For example, some felt stimulated, as affiliated workers, by contact with freelancers: "*Me, having always worked as an employee, getting to know the self-employed world, the freelance scene, it really got me pumped up"* (Luke, CW_FF). This was also described as an advantage for companies that could "*open up to what's going on around them"* (Daniel, CW_CC).

> It is nice above all because coworking is a shared environment with different people doing different jobs. It is not the classic office with the office manager who controls you, where the first thought in the morning is "Oh no I have to go to that place"… I've been an employee for 15 years; I know I'm still an employee, but it's a different situation, isn't it? (Andrew, CW_AA)

Participation in the community and its events to fulfil personal needs. Participation in the community and events related to the coworking space emerged as another advantage for a few interviewees who attended coworking events animated by initiatives and programs concerning their extrawork lives. These interviewees cited events that they considered useful or interesting for their personal lives, such as social and cultural initiatives (e.g., photographic exhibitions), sports, classes, dinners, or breakfasts.

> A main advantage is the participation in events organized by the coworking space, which are similar to its philosophy and have practical advantages, such as the purchase group of local producers that sends food to the coworking space, which is managed by the community manager. (Ricky, CW_EE)

3.2 Building relationships

Building relationships describes the interviewees' relational and support dynamics regarding remote colleagues and coworkers in the coworking space. The following paragraphs detail how the interviewees described the relationships and support mechanisms between remote workers and the social dynamics of the coworking spaces. These themes appeared to be somewhat interconnected since for some individuals, the potential relationships in coworking could compensate for limitations in remote

relationships with colleagues. Meanwhile, others sought to maintain ties with their own company while engaging in coworking.

3.2.1 Challenges in remote collaboration with colleagues: maintaining relationships and overcoming professional isolation

Informal and work relationships with colleagues working outside the coworking space appeared to be perceived differently among the interviewees, highlighting the challenges of remote collaboration. While the frequency and modality of communications may not be a problem for many, remote communication becomes a limitation when participants need to address more complex work issues and relational closeness. Most reported that they communicate daily or frequently with colleagues for work reasons via telephone, chat, or video calls. Especially in cases of work with low levels of interdependence, the participants did not express communication difficulties. In contrast, remote communication is limiting when they face complex or urgent problems. Most importantly, when they spoke about the informal dimension of their relationships, the interviewees described "online relationships" as missing something, and some complained about the loss of more friendly or casual communication with their peers. Some noted that they did not even know some colleagues on their team because of COVID-19. The sense of belonging to the team varied among the interviewees and was a crucial factor for some. Where the relationships were described as positive and there was a sense of belonging to one's work group, the work sphere still seemed to prevail for some, in contrast with the relationships built with fellow coworking space users, which were considered more intimate.

> No, no, no, absolutely not. No, no, I feel what I represent every day because, especially if I need to deal with them, it's a global company, so you always have a response. You can even ask during the day if there's something urgent. However, they are two different relationships in the sense that a real friendship has developed here, not just a professional one. In other words, I do my job and you do yours, we go out and grab a pizza together. With Swiss colleagues, there's a relationship of mutual respect, but primarily it's work-related and not comparable. (Giulie, CW_EE)

Luke found that social connections in coworking spaces satisfied his need for workplace-related social interaction.

> I've always leaned towards self-sufficiency, so I generally enjoy managing myself and doing things independently. Of course, we're social beings, right? So, it's nice to be around… and maybe the fact that I'm with other people here, friends and all, satisfies that social need inside me. I don't really need those moments with colleagues here. (Luke, CW_FF)

Others felt or had felt isolated in the past in terms of working with their teams. This was the case for Daniel, who worked for a foreign company. He travelled to their office every two or three months and felt like a *"mere nobody"* when meeting his col-

leagues. In fact, one interviewee recognized the risk of fragmentation in work groups between those who worked regularly at the office and those who did not. Among the factors that affect social relationships were the perceived richness of ICT utilized and the frequency of face-to-face meetings with the opportunity for more informal exchanges. Virtual cafes, when promoted by the company, are not perceived as being very spontaneous. Instead, workers reported using the moments before or after the start of a web conference to chat more informally. In contrast, one participant mentioned that his company's adoption of an open calendar and the possibility of comanaging video calls with colleagues facilitated a sense of belonging. He highlighted the role of organizational practices that supported social interactions among colleagues, even considering the relationships that could be formed in coworking spaces.

> I currently spend a lot of time on video calls with my colleagues, and this gives me a sense of belonging; I do not feel alone, not at all. Using chat, as I did before, I felt truly alone at the end of the day, regardless of having achieved my work goals. Now, if I need to talk to someone, I am used to, for example, having half-hour virtual coffee chats with colleagues that I don't know… There is this tool for getting to know someone where you have to put a call on his or her calendar… even my colleagues who need to ask me something block out 30 min on my calendar, and my calendar is visible to everyone. This thing here is mind-blowing. I do not manage my calendar alone; I manage it with all my colleagues. (Peter, CW_EE)

More specifically, from the interviewees, it emerged that relationships with their colleagues and companies were facilitated, in part, by face-to-face meetings. Scheduling these meetings within the work group to tackle work projects or at the company headquarters or other locations where affiliated workers could meet, such as "team building" weekends, was seen as a way to both strengthen informal relationships with their colleagues and address work issues.

> Generally, we always had weeks when we would gather together at the headquarters to create a moment where the entire company was united. In my opinion, this is always a positive experience. Once you get to know your colleagues and spend a week working together in the office, it creates a bond. Then, perhaps after six months, you come back for another week, and during that time, you go out for dinner together every night, have a beer, and engage in more personal conversations. Chatting helps to maintain the connections that are built when you meet in person. At least, this is true for me. (Frances, CW_DD)

The absence of planned days at the company headquarters was in fact experienced as a critical issue by one participant, with the expectation that the company should regularly organize such days.

3.2.2 Type of relationships in coworking spaces: from chitchat to support

As discussed earlier, the relational element within the coworking space was highly valued by the interviewees because it fulfils relational social needs that are not satisfied while working from home. Otherwise, the respondents described their relationships with coworkers as having different degrees of intimacy and support. We first describe the forms of relationships and, in the next paragraph, how collaboration and support were experienced. Therefore, we categorize these relationships as *limited-knowledge chitchat*, *class groups*, and *friendships*, as explained below. Some differences depended on multiple factors, such as the time spent in the coworking space, the coworking culture and the type of affiliation, such as whether or not workers were employees of the same company. The interviewees generally described the coworking environment as a place to meet people, although they acknowledged that not all of their coworkers were likely interested in the social aspect. Most described having opportunities to chat and meet with others, but when there was no sense of belonging to an open community, relationships appeared to be limited to "chitchat" and were described as more superficial. Very few interviewees, some of whom had colleagues from the same employer present in the coworking space, classified their relationships as entailing *limited knowledge* of coworkers; these relationships were either reduced to chatting or characterized by closeness with just one person. Beyond the time spent on coworking and personal characteristics, two interesting factors emerged for the affiliated individuals. These interviewees referred to a dynamic of "closure" between colleagues of the same company and the other coworkers that seemed to hamper relations.

> There are many young guys, so obviously in that case, they tend to stay together… that is, there is a group dynamic in which colleagues, obviously, are on their own… but it also applies to chatting with random people. (Camilla, CW_BB)
> Maybe you find us at the restaurant when you go to eat at noon; you may be next to them, but it is always something, for example, *my* company and *your* company. (Daniel, CW_CC)

In addition, unlike for freelancers or start-uppers, building a strong network did not appear particularly relevant for an affiliated worker interviewed:

> I wasn't even in the mode of having to sell my idea and therefore not needing to network with people. No, at the corporate level, I didn't care at all, and even less so on a personal level. (Peter, CW_EE)

The coworkers who used a coworking space that seemed to be characterized by a community-based culture said that they had developed positive relationships with coworkers over time. They tended to share breaks to detach from work issues and chat about personal matters. In particular, many talked about sharing moments within small groups distinguished by greater reciprocity, constancy, and rituality. They compared these groups to *class groups* in high school.

> However, there are people with the same working hours, so you move around a bit together, you go to work together, you leave work together, maybe you go to eat together at the canteen, so it has created for itself a sort of community, a class, and it is something that helps me a lot because when I was at home, I did not see anyone. (Frances, CW_DD)

In addition, these participants referred to widespread knowledge among the stable members of the coworking space, which, for example, manifested in the habit of celebrating birthdays. In this case, different startup teams were described as open and having the habit of "mixing" with each other.

Other interviewees, most of whom used the same coworking space, described relationships that were also pursued outside of the space, explicitly defining these relationships as "*friendships*". They described some of their coworkers as close friends with whom they share the most private topics and spend time outside the workplace, from holidays to evening events, which are planned using group chats.

Collaborative relationships. Experiencing closer relationships appeared to be linked to the perception of support for both work activities and more personal matters. At the same time, coworkers' affiliations with different organizations make support mechanisms flexible and appear to require the management of certain critical aspects relying on commitment to their organization and confidentiality. Some interviewees positively described the *emotional support* provided by coworking members: Because you can see a person when maybe they're a bit more upset than usual, you say to them, "What happened?" "No, look, they are… you know? They are getting on my nerves, the client is upset…" and you talk, exchange opinions. (Andrew_CWAA)

A dozen interviewees described a general openness to the exchange of *practical support* and *work advice*. Asking for and receiving support almost as if "*they were colleagues*" were frequent occurrences with those with closely positioned desks or who were recognized as having the necessary professional skills.

> Ah, there is one on the agenda, of course, but they also ask me. There is this guy M. as well; he's behind me saying, "I'm going crazy over this thing, what can I do?" and then I go check it, and he does the same for me again. It is a good thing. There are all types of professions, so whatever you must do; for example, two or three years ago I had to make videos for conferences, I had to do video editing, and I asked who does it here? (Max, CW_EE)

In addition, support appears to be characterized by not only reciprocity but also freedom from obligations compared to the traditional office situation. In this sense, the interviewees said that they used to offer help when they had time, in contrast to what could occur with office colleagues. Instead, those who had colleagues in the coworking space found it more natural to refer to them for help.

In some cases, participants described opportunities for *work collaboration* among coworkers. While two participants seemed to exclude work collaborations with coworking users, six provided examples of such joint activities. Specifically, collaboration was mentioned as occurring between startups or professionals with perceived similar or complementary competencies. While some expressed satisfaction

with the results of these collaborations for their organization, others had experienced or were planning to engage in independent collaborations and thinking of leaving their companies. Additionally, one participant discussed disadvantages from a company's point of view.

> I can also see some risks, in the sense that you could pass sensitive information on to others, you could let yourself be convinced by someone and then maybe you could change your job… (Daniel, CW_CC).

3.2.3 Sustaining relationships: developing a sense of belonging and a people- and community-based culture

Some interviewees experienced a strong sense of belonging and participation in their coworking space, which is understood as a place that enhances relationships between people. This sense of belonging was perceived to sustain collaborative and supportive relationships, break the boundaries of affiliation, and overcome differences. These interviewees valued the *people and community-oriented culture* of their coworking space, which appeared to be the opposite of a *business-oriented culture.* In the latter type of culture, experienced currently by one respondent and by others in the past, the rules of social coexistence are more rigid, and exchanges are not encouraged.

> Well, the relationship with others is very strong. I do not know if relationships are established as strong as friendships in all coworking spaces like they are here… it is precisely the relationship with others that you feel here is very strong. I visited another one before, and I must tell you there I did not feel at ease; I do not know, the type of coworking space was different. (Giulie, CW_EE)
> Yes, it is definitely different, first of all, because EE is oriented, that is, it is the culture behind it that is different, in the sense that… EE is managed as a community, rather than as a place to have a desk. (Peter, CW_EE)

The sense of belonging to their coworking space was also described as "*living as a family*", thereby qualifying the relationships as open, supportive and based on trust, as Andrew evoked in this excerpt:

> You live in a big family, with different realities; everyone has their own job. Everyone has their own things, of course, but sharing the same roof… in the meantime, we talk to each other… then yes… then something can lead to a collaboration. (Andrew, CW_AA)

Other interviewees positively discussed the "*relaxed"* climate that favours informal relationships, i.e., "*the possibility of interaction in a slightly softer way*", as opposed to the rigidity of the traditional office where relationships are hierarchical. The presence of different professionals was associated with a more dynamic environment that in a certain perception could fuel the climate of the company located in the CW,

allowing members to experience freer relationships with their colleagues than they would in a traditional office. Among the other elements of coworking space that were mentioned as supporting relationships were open spaces, relaxation rooms, dining areas, booksharing, and small coworking spaces. One interviewee talked about table football as a means of becoming acquainted with coworkers; in contrast, a member of a "business" coworking space complained about the absence of larger shared spaces. Furthermore, the distribution of desks was recognized as useful for encouraging collaboration among people with similar jobs. In fact, respondents described having closer relationships with their desk neighbours.

> Then, for example, for me, the playful moments experienced within the coworking space have always been fundamental. So table football, as usually seen in a startup stereotype, was actually for me the only moment of leisure during my lunch break, which is when I connected with the other guys from the other startups. (Stephan, CW_II)

A key role was attributed to the community manager in sustaining *a culture of sharing and mutual support*. For example, one interviewee described the effort made by a coworking manager to respond to the need for childcare for some mothers in the first phase of the COVID-19 pandemic, which led to the creation of a temporary kindergarten. These individuals were defined as "*connectors*", and participants valued the role of the coworking manager in concretely facilitating relationships by putting coworkers in contact with each other. In particular, some interviewees mentioned the welcoming actions performed by the community manager when new members join the coworking space. These managers were seen as facilitators, as they introduced new members to the existing community, organized presentations, and made announcements through social networks.

4 Discussion

This study explores the perceptions of the advantages and disadvantages of coworking spaces for affiliated workers, with a particular focus on relational dynamics. This research seeks to contribute to the literature by identifying the specific experiences of affiliated and remote workers in coworking spaces and examining how affiliation with their own organization and organizational practices can influence these experiences.

Consistent with previous literature considering the overall evaluation of working experience, our findings demonstrate that coworking space can be a supportive work environment and an inclusive solution. In addition to previous research, this study highlights some specific key conditions for affiliated workers to make coworking spaces places where they can have more or less social relationships, reduce isolation, and engage in collaboration (Lescarret et al. 2022).

The results show that coworking is a supportive solution, identifying two major advantages of coworking space: as supporting work–life balance and as a catalyst for relationships. Simultaneously, the analysis shows that these advantages are prioritized differently among respondents, aligning partly with previous research that

categorized coworking users into two business models: those seeking efficiency and those seeking novelty (Bouncken et al. 2020).

Furthermore, the study explores the dynamics of coworking relationships considering the mix of affiliated and unaffiliated workers and highlights variations in social collaboration based on the quality of organizational identification with the affiliated organization and professional identity (Ashforth et al 2000) . The findings indicate that while coworking can facilitate relationships with internal colleagues, it may not necessarily lead to deeper connections with other coworkers. The social dynamics of coworking may change for groups of different categories of workers (affiliated workers and freelancers), who can perceive themselves as distinct and experience the space differently (Rodighiero 2023). This enhances the understanding of coworking space as a multifaceted social setting, thereby contributing to a nuanced interpretation of the coworking experience.

With regard to "coworking as a flexible and efficient solution", coworking spaces appear to help reduce the interference between private life and work, promote flexibility, and sustain a sense of productivity and well-being. The results also indicate that the perception of greater autonomy, greater flexibility, and, importantly, less control becomes a positive aspect of working in a coworking space compared to a company office. This is linked to commuting time savings and the perception of greater flexibility in working hours and alternation with working from home.

It should be noted that while companies often promote coworking spaces as an inclusive and flexible solution, paying office rent is rare in our sample, excluding companies based in coworking spaces. According to social exchange theory (Blau 2017), this organizational support can preserve affiliated workers' organizational commitment (Bodjrenou et al 2019). However, based on our data, companies only partially meet their workers' need for flexibility, mainly through flexible schedules and the absence of control devices. Indeed, we find that a lack of support for the use of coworking spaces can be a source of dissatisfaction for some remote workers.

In terms of the social dimension, coworking is perceived as not only as a positive environment for informal socialization but also a place for some interviewees to experience support, collaboration, and a different working culture than in traditional office settings. More rarely, satisfaction is expressly connected to participation in the community and events.

The literature has already highlighted the presence of two configurations: coworkers as "good neighbours" who work alone and focus on their tasks and coworkers as "good partners" who actively interact to establish formal collaboration (Spinuzzi 2012). In the case of affiliated workers, to explain these differences, we can also refer to their relationships with their company and the definition or redefinition of their professional identity. First, where the connection with one's organization is weaker due to physical distance or a sense of isolation, it seems possible to develop institutional patterns of linkage multiplicity and mutual knowledge creation (Bouncken et al. 2020). At the same time, these institutional patterns may be prevented because some affiliated workers experience requests for work support from coworkers as flexible. This perception is attributed to the absence of formal or informal collaboration constraints, distinguishing it from the dynamics observed in traditional workplaces with colleagues from the same team.

For instance, for remote workers experiencing a sense of isolation, coworking becomes a space to reinforce a sense of professional identity and belonging to a community. It is possible, therefore, to experience work support and collaboration even with coworkers who are not affiliated.

Coworking also serves as a space where professionals can redefine their professional identity through interactions with individuals from diverse backgrounds. Professional exchange is motivated not only by instrumental values but also by the perception of learning opportunities based on different professional identities.

For some, especially those with internal colleagues in coworking spaces, the bond with their company appears prevalent, and relational dynamics seem more superficial and less focused on support with other coworkers.

For others, the motivation for reduced socialization lies in their professional identity and perceived absence of professional and economic benefits, which are often associated with freelancers or startups (Bouncken and Reuschl 2018) 2020. Additionally, collaboration requires the management of certain critical aspects that rely on commitment to their organization and confidentiality.

In line with the current literature, a key factor is the coworking environment and the different "community potential" (Parrino 2015; Berdicchia et al. 2023), which can define and redefine relational hierarchies. Copresence in coworking spaces does not ensure supportive or even collaborative relationships (Ivaldi et al. 2022). Rather, a climate of collaboration and the dissemination of community-based values nurture mutual support practices. When cooperative values are promoted and a sense of belonging is developed, users are encouraged to share their competencies, experiences, and knowledge (Bouncken and Aslam 2019), thereby making the boundaries of affiliations among workers permeable. The physical aspects and arrangement of workstations in open spaces are confirmed as key elements supporting sociability, despite some disadvantages, such as noise, interruptions, lack of privacy, and data protection concerns (Hurry 2012; Robelski et al. 2019; Josef 2020).

Consequently, our results illustrate that social support is experienced among coworkers, demonstrating greater flexibility than in relationships within the same organization. This perceived flexibility enables affiliated workers to implement boundary behaviours, preserving their concentration on work when needed and reducing interruptions, which are considered counterproductive.

This study highlights the importance of professional exchange and collaboration in coworking, even among unaffiliated coworkers. This underscores the role of coworking spaces as environments that foster diverse professional identities and learning opportunities, contributing to the literature on the collaborative aspects of coworking spaces.

In addition to these insights into the inclusive value of coworking for remote workers, the results show that the organization has a role in managing critical aspects. The loosening of organizational boundaries has the potential to disrupt organizational communication and the relationship between affiliated workers and the organization. The copresence of workers from different working realities paves the way for reduced control over company information and threatens organizational commitment if coworkers find more attractive job opportunities.

There is an expressed need for effective management of remote work, considering the need for interaction and socialization among colleagues in a virtual environment where social presence is reduced (Fonner and Roloff 2012). To mitigate professional isolation, the findings underscore the significance of organizational practices, including periodic face-to-face meetings and communication strategies that facilitate more robust forms of remote interaction (Golden et al. 2008; Van Zoonen and Sivunen 2022).

The findings suggest that while companies may promote coworking as an inclusive solution, there is a gap between rhetoric and practice. This finding sheds light on the complexities of organizational support in the context of coworking and digitization of work (Chopra et al. 2024).

5 Limitations and future directions

Our study has some limitations that may serve as the basis for future research. Our methodology led to the construction of a varied sample in terms of remote working conditions, uses, and types of coworking spaces, which does not allow us to clearly distinguish their influence on the perception of the advantages of coworking spaces. Thus, it could be useful to conduct a comparative case study of different coworking spaces selected by their characteristics to expand the understanding of the impact of the spaces' organizational characteristics (culture, size, spaces) and the type of affiliation (freelancer or employee working remotely) on social relationships. Including freelancers in the sample allowed us to compare the social dynamics of individual workers and in teams in coworking spaces with those of affiliated workers and to better understand the differences between freelancers and remote workers. Furthermore, our study did not consider the views and practices of corporate managers responsible for supervising workers or teams in coworking spaces. Finally, our sample did not include affiliated workers of public companies, which have increasingly adopted telework in recent years. Future research should also consider this category of workers.

6 Conclusion and recommendations for practice

The current study discusses the advantages of the use of coworking space and its implications for work practices, offering insight for companies considering affiliated workers' perceptions. From a corporate management perspective, the identified perceived advantages and preferences can help determine which workplace will be most comfortable for affiliated workers.

Coworking space can be a solution, as an alternative to the home office, for those remote workers who have a high relational need at work (Deci and Ryan 1985) and prefer segmentation between the working and domestic spheres (Kreiner 2006). Coworking space can support workers' perception of flexibility, which is valued by affiliated workers, if rigid work hours are not imposed and if workers are supported through the rental of workstations.

Moreover, if work–life balance is considered more comprehensively—that is, including the life domains of household, relationship, health, family, etc. (Keeney et al. 2013)—the use of coworking space can help achieve it when workers participate in activities concerning parenting, household management, leisure, and friendship (Orel and Alonso Almeida 2019; Berbegal-Mirabent 2021).

Coworking space can be an inclusive choice if management carefully chooses the space considering its physical, social, and cultural aspects. We discussed how copresence in coworking spaces does not ensure building supportive or even collaborative relationships but is essential for a community-based culture. Managers should help their affiliated workers find a suitable workspace and engage in supportive communication with their remote workers to empower them to collaborate with outsiders under win–win conditions.

Furthermore, due to the opening of organizational boundaries, exchanges between diverse professionals become learning opportunities at both the individual and organizational levels. The work culture of some coworking spaces, which is based on less hierarchical relationships, could bring additional advantages to contemporary organizations (Josserand et al. 2006).

Collaboration and support can be sustained when workers have time or when they see a clear benefit. In addition, organizations should establish clear corporate confidentiality policies and help their affiliated workers to know whether information should be considered confidential.

Moreover, in addition to the use of coworking spaces, management should adopt adequate technological infrastructure and policies to support the perception of social presence and organizational belonging. The findings suggest that individuals struggle with tasks and risk feeling isolated when they are physically separated from their colleagues. Facilitating encounters by leaving open time both before and after online meetings or by organizing informal encounters involving online social events could also allow informal communication (Larson et al. 2020). Thus, organizational identification is achieved mainly through communication with others (Parker and Haridakis 2008).

The results of the current study therefore push us to extend the current models of virtual leadership (Avolio et al. 2000) for dispersed teams, considering the specificities of the coworking environment. Managers are called not only to support relationships between team members but also to support workers in finding ways to benefit from support and collaboration with coworkers within the same coworking space.

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Data availability The data that support the findings of this study are available from the corresponding author upon request.

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References

Appel-Meulenbroek R, Weijs-Perrée M, Orel M et al (2021) User preferences for coworking spaces; a comparison between the Netherlands, Germany and the Czech Republic. RMS 15:2025–2048. https://doi.org/10.1007/s11846-020-00414-z

April CNBC (2021) 23, The new negotiation over job benefits and perks in post-Covid hybrid work. Deskmag (2022). 2021-22 Coworking Space Trends. https://coworkingstatistics.com/coworkingstatistics/2021-22-coworking-space-trends

Ashforth BE, Mael F (1989) Social identity theory and the organization. Acad Manage Rev 14(1):20–39. https://doi.org/10.2307/258189

Ashforth BE, Kreiner GE, Fugate M (2000) All in a day's work: boundaries and Micro Role transitions. Acad Manage Rev 25:472. https://doi.org/10.2307/259305

Avolio BJ, Kahai S, Dodge GE (2000) E-leadership. Leadersh Q 11:615–668. https://doi.org/10.1016/S1048-9843(00)00062-X

Barbour RS (2001) Checklists for improving rigour in qualitative research: a case of the tail wagging the dog? Br Med J 322:1115. https://doi.org/10.1136/bmj.322.7294.1115

Bartel CA, Wrzesniewski A, Wiesenfeld BM (2012) Knowing where you stand: physical isolation, perceived respect, and Organizational Identification among virtual employees. Organ Sci 23:743–757. https://doi.org/10.1287/orsc.1110.0661

Berbegal-Mirabent J (2021) What do we know about Co-working spaces? Trends and challenges ahead. Sustainability 13:1416. https://doi.org/10.3390/su13031416

Berdicchia D, Fortezza F, Masino G (2023) The key to happiness in collaborative workplaces. Evidence from coworking spaces. RMS 17:1213–1242. https://doi.org/10.1007/s11846-022-00558-0

Bianchi F, Casnici N, Squazzoni F (2018) Solidarity as a byproduct of professional collaboration: social support and trust in a coworking space. Soc Networks 54:61–72. https://doi.org/10.1016/j.socnet.2017.12.002

Bilandzic M, Foth M (2013) Libraries as coworking spaces. Libr Hi Tech 31:254–273. https://doi.org/10.1108/07378831311329040

Blau PM (2017) Exchange and Power in Social Life. Routledge

Bodjrenou K, Xu M, Bomboma K (2019) Antecedents of Organizational Commitment: a review of personal and organizational factors. Open J Soc Sci 07:276–289. https://doi.org/10.4236/jss.2019.75024

Bouncken RB, Reuschl AJ (2018) Coworking-spaces: how a phenomenon of the sharing economy builds a novel trend for the workplace and for entrepreneurship. RMS 12:317–334. https://doi.org/10.1007/s11846-016-0215-y

Bouncken R, Aslam, MM (2019) Understanding knowledge exchange processes among diverse users of coworking-spaces. J. Knowl. Manag 23 (10): 2067-2085. https://doi.org/10.1108/JKM-05-2018-0316

Bouncken RB, Ratzmann M, Barwinski R, Kraus S (2020) Coworking spaces: empowerment for entrepreneurship and innovation in the digital and sharing economy. J Bus Res 114. https://doi.org/10.1016/j.jbusres.2020.03.033

Bouncken RB, Aslam MM, Qiu Y (2021) Coworking spaces: understanding, using, and managing sociomateriality. Bus Horiz 64:119–130. https://doi.org/10.1016/j.bushor.2020.09.010

Braun V, Clarke V (2006) Using thematic analysis in psychology. Qual Res Psychol 3:77–101. https://doi.org/10.1191/1478088706qp063oa

Bréchignac B (2016) Corporate coworking: what is the reality behind it? https://collaborativespacesstudy.wordpress.com/2016/01/30/corporate-coworking-what-is-the-reality-behind-it/

Brinks V (2022) Fluid objects? An attempt to conceptualise the global rise of coworking spaces. Area 54:435–442. https://doi.org/10.1111/area.12778

Bueno S, Rodríguez-Baltanás G, Gallego MD (2018) Coworking spaces: a new way of achieving productivity. J Facilities Manage 16:452–466. https://doi.org/10.1108/JFM-01-2018-0006

Byron K (2005) A meta-analytic review of work–family conflict and its antecedents. J Vocat Behav 67:169–198. https://doi.org/10.1016/j.jvb.2004.08.009

Capdevila I (2013) Knowledge dynamics in localized communities: Coworking spaces as Microclusters. SSRN Electron J 1–18. https://doi.org/10.2139/ssrn.2414121

Chopra R, Agrawal A, Sharma GD et al (2024) Uncovering the organizational, environmental, and socio-economic sustainability of digitization: evidence from existing research. Rev Manag Sci 18:685–709. https://doi.org/10.1007/s11846-023-00637-w

Collins AM, Hislop D, Cartwright S (2016) Social support in the workplace between teleworkers, office-based colleagues and supervisors. New Technol Work Employ 31:161–175. https://doi.org/10.1111/ntwe.12065

Corbin J, Strauss A (2008) Basics of Qualitative Research (3rd ed.): Techniques and Procedures for Developing Grounded Theory. SAGE Publications, Inc., 2455 Teller Road, Thousand Oaks California 91320 United States

Criscuolo C, Gal P, Leidecker T et al (2022) The role of Telework for Productivity during and post-covid-19. Results From an Oecd Survey Among Managers and Workers

Deci EL, Ryan RM (1985) The general causality orientations scale: self-determination in personality. J Res Pers 19:109–134. https://doi.org/10.1016/0092-6566(85)90023-6

Deskmag (2013) The third global coworking survey 2012. https://www.deskmag.com/en/coworking-news/1st-results-of-the-3rd-global-coworking-survey-2012

DiMaggio PJ, Powell WW (1983) The iron cage revisited: Institutional isomorphism and collective rationality in organizational fields. Am. Sociol. Rev 48(2): 147-160.

Foertsch C (2019) State of Coworking: Over 2 Million Coworking Space Members Expected. https://www.deskmag.com/en/coworking-news/2019-state-of-coworking-spaces-2-million-members-growth-crisis-market-report-survey-study

Foertsch C (2020) Coworking Space Members: It's a girl! https://www.deskmag.com/en/coworking-city-country-profiles/coworking-space-members-demographics-market-report-study-survey-1034

Fonner KL, Roloff ME (2012) Testing the Connectivity Paradox: linking Teleworkers' Communication Media Use to Social Presence, stress from interruptions, and Organizational Identification. Commun Monogr 79:205–231. https://doi.org/10.1080/03637751.2012.673000

Gandini A (2015) The rise of coworking spaces: a literature review. Ephemera: Theory Politics Organ 15(1):193–205

Garrett LE, Spreitzer GM, Bacevice PA (2017) Co-constructing a sense of community at work: the emergence of community in Coworking spaces. Organ Stud 38:821–842. https://doi.org/10.1177/0170840616685354

Gerdenitsch C, Scheel TE, Andorfer J, Korunka C (2016) Coworking spaces: a source of Social Support for Independent professionals. Front Psychol 7. https://doi.org/10.3389/fpsyg.2016.00581

Golden TD, Veiga JF, Dino RN (2008) The impact of professional isolation on teleworker job performance and turnover intentions: does time spent teleworking, interacting face-to-face, or having access to communication-enhancing technology matter? J Appl Psychol 93:1412–1421. https://doi.org/10.1037/a0012722

Halford S (2005) Hybrid workspace: re-spatialisations of work, organisation and management. New Technol Work Employ 20:19–33. https://doi.org/10.1111/j.1468-005X.2005.00141.x

Horton KE, Bayerl PS, Jacobs G (2014) Identity conflicts at work: an integrative framework. J Organ Behav 35:6–22. https://doi.org/10.1002/job.1893

Howell T (2022) Coworking spaces: an overview and research agenda. Res Policy 51:104447. https://doi.org/10.1016/j.respol.2021.104447

Hurry CJP (2012) The HUB Halifax: A Qualitative Study on Coworking

Italian Coworking Survey (2021) (2022). Indagine su coworking e uffici flessibili. Italian Coworking https://www.italiancoworking.it/italian-coworking-survey-2021-dati-vizzes-e-reports/

Ivaldi S, Sannino A, Scaratti G (2022) Is 'co' in coworking a short for contradictions? Qualitative Research in Organizations and Management. Int J 17:38–63. https://doi.org/10.1108/QROM-06-2020-1970

Jeske D, Ruwe T (2019) Inclusion through use and membership of co-working spaces. J Work-Applied Manage 11:174–186. https://doi.org/10.1108/JWAM-06-2019-0021

Josef B (2017) Coworking from the Company's Perspective - Serendipity-biotope or Getaway-spot? In: 30th Bled eConference: Digital Transformation - From Connecting Things to Transforming our Lives, BLED 2017. Association for Information Systems Electronic Library, AISeL, pp 263–278
Josef BD (2020) Coworking from a Corporate Perspective-out of Office, into the Flow? Doctoral dissertation, Universität St. Gallen
Josserand E, Teo S, Clegg S (2006) From bureaucratic to post-bureaucratic: the difficulties of transition. J Organizational Change Manage 19:54–64. https://doi.org/10.1108/09534810610643686
Keeney J, Boyd EM, Sinha R et al (2013) From work–family to work–life: broadening our conceptualization and measurement. J Vocat Behav 82:221–237. https://doi.org/10.1016/j.jvb.2013.01.005
Ker D, Montagnier P, Spiezia V (2021) Measuring telework in the COVID-19 pandemic. OECD Digital Economy Papers No. 314
Kojo I, Nenonen S (2016) Typologies for co-working spaces in Finland – what and how? Facilities 34:302–313. https://doi.org/10.1108/F-08-2014-0066
Kraus S, Bouncken RB, Görmar L, González-Serrano MH, Calabuig F (2022) Coworking spaces and makerspaces: mapping the state of research. J Innov Knowl 7:100161. https://doi.org/10.1016/j.jik.2022.100161
Kraus S, Ferraris A, Bertello A (2023) The future of work: how innovation and digitalization re-shape the workplace. J Innov Knowl 8:100438. https://doi.org/10.1016/j.jik.2023.100438
Kreiner GE (2006) Consequences of work-home segmentation or integration: a person-environment fit perspective. J Organ Behav 27:485–507. https://doi.org/10.1002/job.386
Larson BZ, Vroman SR, Makarius EE (2020) March 18). A guide to managing your (newly) remote workers. Harvard Business Rev https://hbr.org/2020/03/a-guide-to-managing-your-newly-remote-worker
Leclercq-Vandelannoitte A (2021) Do coworking spaces promise a revolution or spark revenge? A foucauldian Spatio-Material Approach to the re-spatialization of remote work in Coworking spaces. 151–173. https://doi.org/10.1007/978-3-030-61687-8_7
Leclercq-Vandelannoitte A, Isaac H (2016) Journal of Business Strategy the new office: how coworking changes the work concept. J Bus Strategy J Bus Strategy Iss J Bus Strategy Iss 37:3–9. https://doi.org/10.1108/JBS-10-2015-0105
Lescarret C, Lemercier C, Le Floch V (2022) Coworking spaces vs. home: does employees' experience of the negative aspects of working from home predict their intention to telework in a coworking space? Front Psychol 13:1079691. https://doi.org/10.3389/fpsyg.2022.1079691
Manzini Ceinar I, Mariotti I (2022) May local coworking spaces be the future trend? Romanian J Reg Sci 15(1):52–76
Miglioretti M, Gragnano A, Simbula S, Perugini M (2023) Telework quality and employee well-being: lessons learned from the COVID-19 pandemic in Italy. New Technol Work Employ 38(3):548–571. https://doi.org/10.1111/ntwe.12263
Mitev N, de Vaujany F-X, Laniray P et al (2019) Co-working spaces. Collaborative Practices Entrepreneurship 15–43. https://doi.org/10.1007/978-3-319-94487-6_2
Oldenburg R, Brissett D (1982) The third place. Qual Sociol 5:265–284. https://doi.org/10.1007/BF00986754
Orel M, Bennis WM (2021) Classifying changes. A taxonomy of contemporary coworking spaces. J Corp Real Estate 23:278–296. https://doi.org/10.1108/JCRE-12-2020-0061
Orel M, del Alonso Almeida M M (2019) The ambience of collaboration in coworking environments. J Corp Real Estate 21:273–289. https://doi.org/10.1108/JCRE-12-2018-0050
Orel M, Mayerhoffer M (2021) Qualitative research methods for the exploration of coworking environments. Qualitative Rep 26(5):1364–1382. https://doi.org/10.46743/2160-3715/2021.4673
Parker R, Haridakis P (2008) Development of an organizational identification scale: integrating cognitive and communicative conceptualizations. J Communication Stud 1:105–246
Parrino L (2015) Coworking: assessing the role of proximity in knowledge exchange. Knowl Manage Res Pract 13:261–271. https://doi.org/10.1057/kmrp.2013.47
Patton MQ (2002) Qualitative research and evaluation methods. Thousand Oaks. Sage, Cal
Rådman E, Johansson E, Bosch-Sijtsema P et al (2023) In search of member needs in coworking spaces. Rev Manag Sci 17:881–907. https://doi.org/10.1007/s11846-022-00546-4
Reuschke D, Clifton N, Fisher M (2021) Coworking in homes–mitigating the tensions of the freelance economy. Geoforum 119:122–132. https://doi.org/10.1016/j.geoforum.2021.01.005
Robelski S, Keller H, Harth V, Mache S (2019) Coworking spaces: the Better Home Office? A Psychosocial and Health-related perspective on an emerging work environment. Int J Environ Res Public Health 16:2379. https://doi.org/10.3390/ijerph16132379

Rodighiero S (2023) Coworking spaces: thriving and organizing in the new world of work. https://doi.org/10.48676/unibo/amsdottorato/11063. Dissertation thesis

Ross S, Ressia (2015) Neither Office nor Home: Coworking as an emerging workplace choice. Employ Relations Record 15:42–57

Sargent K, Cooper J, Mellwig B, McDonald M (2018) Coworking and the disruption of the current corporate real estate model. Corp Real Estate J 7(3):267–276

Schäfer B, Koloch L, Storai D, Gunkel M, Kraus S (2023) Alternative workplace arrangements: tearing down the walls of a conceptual labyrinth. J Innov Knowl 8:100352. https://doi.org/10.1016/j.jik.2023.100352

Servaty R, Perger G, Harth V, Mache S (2018) Working in a cocoon: (Co)working conditions of office nomads - a health related qualitative study of shared working environments. Work 60:527–538. https://doi.org/10.3233/WOR-182760

Spinuzzi C (2012) Working alone together. J Bus Tech Commun 26:399–441. https://doi.org/10.1177/1050651912444070

Spinuzzi C, Bodrožić Z, Scaratti G, Ivaldi S (2019) Coworking is about community: but what is community in Coworking? J Bus Tech Commun 33:112–140. https://doi.org/10.1177/1050651918816357

Spreitzer GM, Garrett LE, Bacevice P (2015) Should your company embrace coworking? MIT Sloan Manag Rev 57. https://sloanreview.mit.edu/article/should-your-company-embrace-coworking/

Spreitzer GM, Bacevice PA, Hendricks H, Garrett LE (2020) Community in the new world of work: implications for organizational development and thriving. In: Noumair DA, Shani AB (eds) Research in organizational change and development. Emerald Publishing, Bingley, UK, pp 77–101

Van Zoonen W, Sivunen AE (2022) The impact of remote work and mediated communication frequency on isolation and psychological distress. Eur J Work Organizational Psychol 31:610–621. https://doi.org/10.1080/1359432X.2021.2002299

Walden J (2019) Communicating role expectations in a coworking office. J Communication Manage 23:316–330. https://doi.org/10.1108/JCOM-09-2018-0097

Waters-Lynch J, Duff C (2021) The affective commons of Coworking. Hum Relat 74:383–404. https://doi.org/10.1177/0018726719894633

Waters-Lynch J, Potts J (2017) The social economy of coworking spaces: a focal point model of coordination. Rev Soc Econ 75:417–433. https://doi.org/10.1080/00346764.2016.1269938

Waters-Lynch JM, Potts J, Butcher T et al (2016) Coworking: a Transdisciplinary overview. SSRN Electron J. https://doi.org/10.2139/ssrn.2712217

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